2018EBN122C2	**2018EBN122C2**
CLASS TEST	**KLASTOETS**
Kopiereg voorbehou	Copyright reserved
21 September 2018	21 September 2018

MEMORANDUM

Maksimum punte: *Maximum marks:*	16	Volpunte: *Full marks:*	15
Duur van vraestel: *Duration of paper:*	30 min + 10 min to complete OCR form *30 min + 10 min om OKH vorm te voltooi*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		8 + OCR	

Important

- Die eksamenregulasies van die Universiteit van Pretoria geld
  The examination regulations of the University of Pretoria apply.
- **All questions must be answered on the Optical Character Recognition (OCR) form.**
  Alle vrae moet op die Optiese Karakter Herkenning ('OCR')- vorm ingevul word.
- Read the instructions on the OCR form carefully.
  Neem noukeurig kennis van die instruksies op die 'OCR'-vorm
- Answers must include units!
  Antwoorde moet eenhede insluit!
- **Final answers must be written as real numbers and not as fractions, and must be rounded to <u>5 significant numbers</u>**
  Finale antwoorde moet as reële getalle geskryf word en nie as breuke nie, en moet afgerond word na <u>5 beduidende syfers</u>.

Dosent(e) / *Lecturer(s):*	D le Roux, JP de Villiers, JP Jacobs

INSTRUCTIONS

Do step 0 in the first 5 minutes of the *session*. (0-5 min)
Do step 1 in the first 30 minutes of the *test*. (5-35 min)
Do step 2 within the following 10 minutes. (35-45 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.
- The Assessment ID is:

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- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	–	1	.	4	5					µ	A								
02	3	4	0							V									
03	1	5	.	2						k									
04	2	0								Ω									

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die *sessie*. (0-5 min)
Doen stap 1 in die eerste 30 minute van die *toets*. (5-35 min)
Doen stap 2 binne die daaropvolgende 10 minute. (35-45 min)

STAP 0 (-5 min to 0 min): Voltooi die voorblad van die *OCR*-vorm

STAP 1 (0 min to 30 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN-ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN-ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (35 min to 40 min): Dra die antwoorde vanaf die OEFEN-ANTWOORDBLAD oor na die *OCR*-vorm.
- Die *Assessment ID* is:

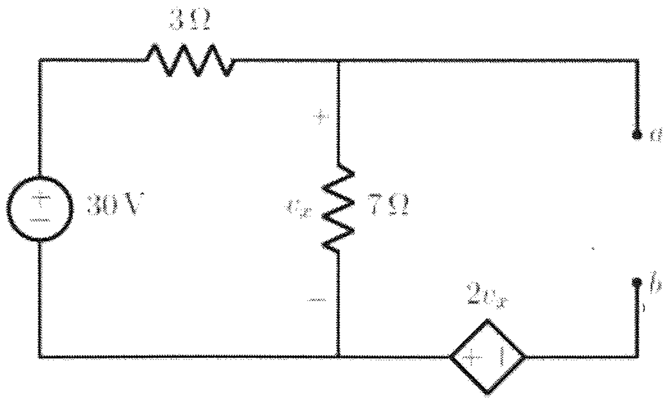
$\boxed{\textbf{2018EBN122C2}}$

- Gebruik 'n meganiese potlood met 0.5mm HB lood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak asb. gebruik van punte.

Voorbeelde

01	–	1	.	4	5					m	A							
02	3	4	0							V								
03	1	5	.	2														
04	-	1								Ω								

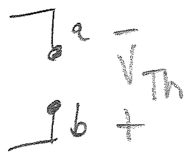
Question 1-2 [4]		Vraag 1-2 [4]	
Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} with reference to terminals *a*-*b*.		Bepaal die Thevenin ekwivalente spanning V_{Th} en weerstand R_{Th} met verwysing tot terminale *a* en *b*.	
01	V_{Th}	**01**	V_{Th}
02	R_{Th}	**02**	R_{Th}



V_{Th}:

KVL: $v_x + V_{Th} + 2v_x = 0$

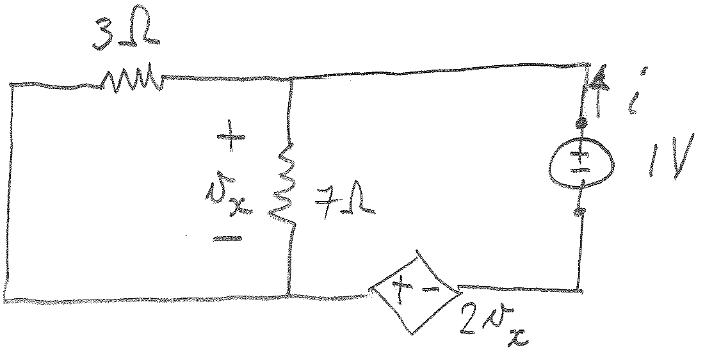
$v_x = 30 \cdot \frac{7}{10} = 21$ V



$\therefore \quad 21 + V_{Th} + 42 = 0$

$V_{Th} = -63$ V

R_{Th}:



$v_x - 1 + 2v_x = 0$

$v_x = i(3 \| 7)$

$= 2.1i$

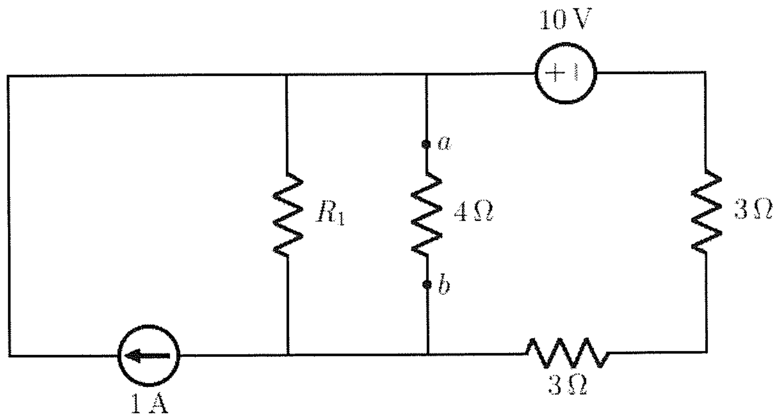
$2.1i - 1 + 4.2i = 0$

$\Rightarrow i = 0.15873$ A

$R_{Th} = \frac{1}{i}$

$= 6.3\ \Omega$

Question 3 [2]		Vraag 3 [2]	
03	It is known that maximum power is delivered to the 4 Ω load resistor. Determine R_1.	03	Dit is bekend dat maksimum drywing oorgedra word aan die 4 Ω lasweerstand. Bepaal R_1.



Max power transfer to 4 Ω implies

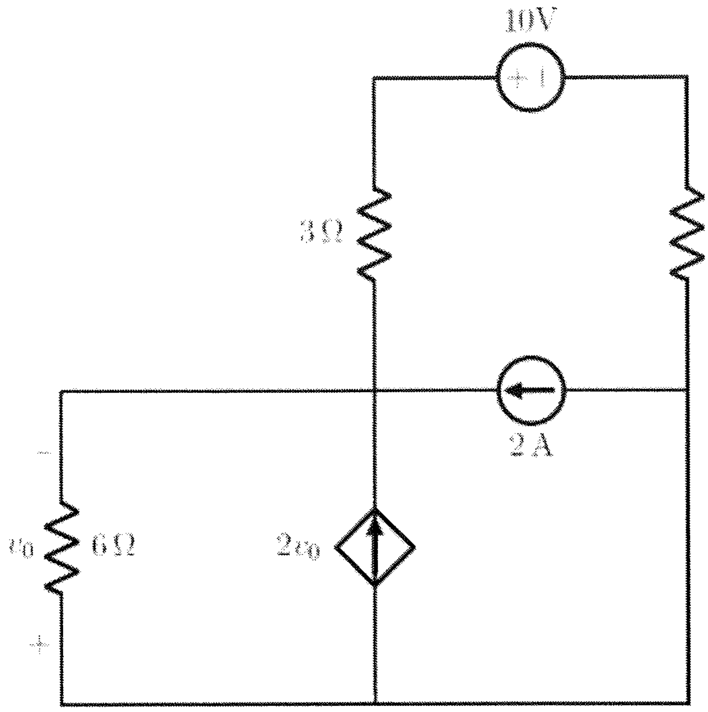
$$R_{Th} = 4\ \Omega$$

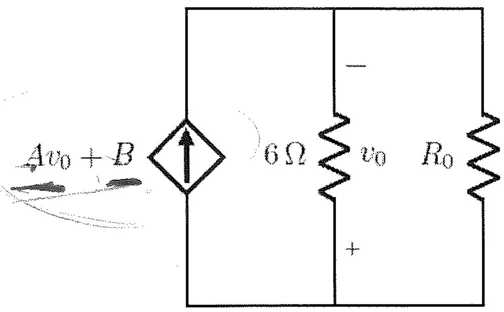
$$R_{Th} = R_1 \parallel (3+3)$$

$$4 = \frac{6R_1}{6+R_1}$$

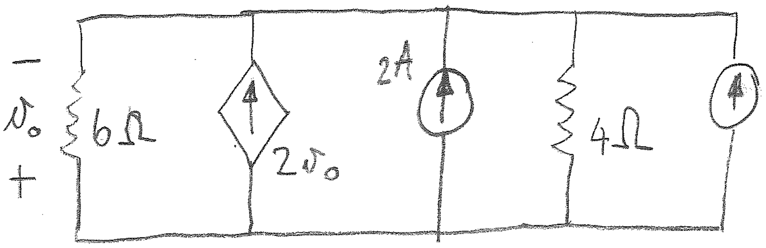
$$24 + 4R = 6R \Rightarrow R_1 = 12\ \Omega$$

Question 4 - 6 [4]		Vraag 4 - 6 [4]	
Simplify the circuit on the left by means of source transformation so that it looks like the one on the right. Determine:		Vereenvoudig die stroombaan aan die linkerkant met behulp van brontransformasie sodat dit lyk soos die kringbaan aan die regterkant. Bepaal:	
04	R_0 [1]	**04**	R_0 [1]
05	A [1]	**05**	A [1]
06	B [2]	**06**	B [2]





Q 4-6



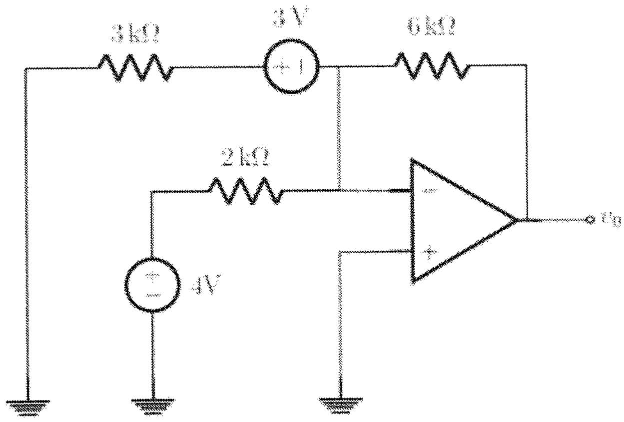
$\frac{10}{4} = 2{,}5$ A.

$\Rightarrow$ $A = 2$

$B = 4.5$ (sum of independent voltage sources)

$R_0 = 4\ \Omega$

Question 7 [2]		Vraag 7 [2]	
07	Determine v_o. Assume that the op-amp is ideal.	07	Bepaal v_o. Aanvaar dat die OV ideaal is.



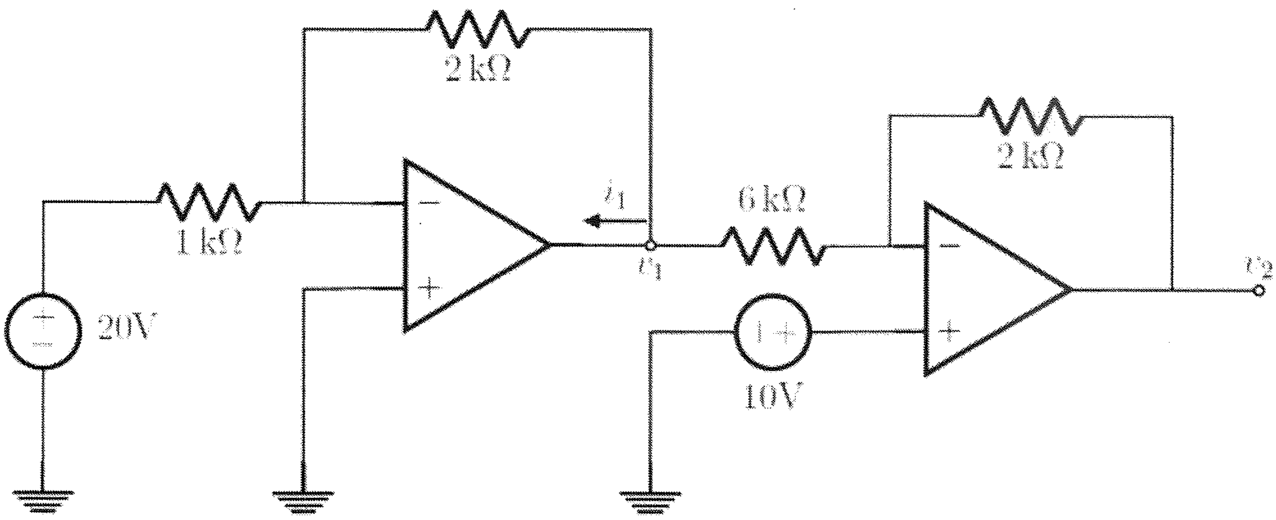
Summing amplifier:

$$v_o = -\left(\frac{6000}{3000}(-3) + \frac{6000}{2000}(4)\right)$$

$$= -(-6 + 12)$$

$$= -6 \text{ V}$$

Questions 8-9 [4]		Vraag 8-9 [4]	
Two ideal op-amps are interconnected as indicated. It is known that $v_1 = -40$ V.		Twee ideale OVs word verbind soos aangetoon. Dit is bekend dat $v_1 = -40$ V.	
08	Determine v_2. [2]	**08**	Bepaal v_2. [2]
09	Determine i_1. [2]	**09**	Bepaal i_1. [2]



Q8-9 $\left(v_1 = -\frac{2000}{1000} 20 = -40 \text{ V}\right)$

v_2: $\frac{10 - v_1}{6000} = \frac{v_2 - 10}{2000}$

$10 - (-40) = 3v_2 - 30 \quad \therefore \; v_2 = \frac{80}{3} = 26.667 \text{ V}$

i_1: $i_1 = \frac{0 - v_1}{2000} + \frac{10 - v_1}{6000}$

$= \frac{0 - (-40)}{2000} + \frac{10 - (-40)}{6000} = 28.333 \text{ mA}$

PRACTICE ANSWER SHEET / OEFEN-ANTWOORDBLAD

This page may NOT be removed from the test. It will not be taken in.

Assessment ID	2	0	1	8	E	B	N	1	2	2	C	2		

Answer / Antwoord

Answers must be rounded to <u>5 significant numbers</u>
Antwoorde moet afgerond word na <u>5 beduidende syfers</u>.

Unit / Eenheid

01	V_{Th} = −63	V
02	R_{Th} = 6.3	Ω
03	R_I = 12	Ω
04	R_0= 4	Ω
05	A = 2	
06	B = 4.5	
07	v_0 = −6	V
08	v_2 = 26.667	V
09	i_1 = 28.333	mA.